

OSLO BLINDERN, Norway

WMO: 014920

Lat: 59.9422N

Lon: 10.7200E

Elev: 318

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	10.9	1.8	6.1	10.0	5.3	7.3	15.1	19.9	32.8	17.8	33.8	3.8	30	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	80.5	63.1	77.0	62.1	73.7	60.3	65.6	75.3	64.0	73.0	62.5	70.9	6.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	84.7	68.5	60.6	79.9	66.9	59.1	75.6	65.7	30.6	75.6	29.4	73.0	28.3	71.0	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.3	13.5	3.0	84.6	4.4	4.7	-0.2	88.0	-2.7	90.7	-5.2	93.3	-8.4	96.7		
(5)			2.1	67.6	4.3	1.8	-1.0	68.8	-3.5	69.9	-5.8	70.9	-8.9	72.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	27.4	28.7	34.5	43.5	52.7	59.6	64.4	62.2	54.5	43.9	35.9	29.2	(6)	(7)
	DBStd	14.60	8.19	7.78	7.05	5.97	6.76	5.07	4.95	4.73	5.03	5.90	6.67	8.34	(7)	(8)
	HDD50	3322	700	596	482	210	45	1	0	0	17	203	424	644	(8)	(9)
	HDD65	7473	1165	1016	946	644	387	177	73	115	314	653	874	1109	(9)	(10)
	CDD50	1424	0	0	1	15	129	288	445	379	153	14	0	0	(10)	(11)
	CDD65	101	0	0	0	0	5	14	53	29	0	0	0	0	(11)	(12)
	CDH74	678	0	0	0	0	43	138	360	136	1	0	0	0	(12)	(13)
	CDH80	118	0	0	0	0	5	23	78	12	0	0	0	0	(13)	(14)
(14) Wind	WSAvg	6.1	5.8	5.7	6.1	6.5	6.7	6.5	5.9	5.8	6.0	5.9	6.2	5.7	(14)	(15)
(15) Precipitation	PrecAvg	30.4	2.0	1.5	1.8	1.7	2.0	2.7	3.2	3.4	3.5	3.3	3.0	2.1	(15)	(16)
	PrecMax	46.2	5.5	3.5	4.8	3.6	4.6	5.6	7.1	9.0	9.0	7.4	11.0	5.4	(16)	(17)
	PrecMin	22.1	0.2	0.0	0.0	0.0	0.0	0.7	0.6	0.5	1.1	0.5	0.4	0.3	(17)	(18)
	PrecStd	5.1	1.3	0.9	1.2	1.0	1.1	1.3	1.5	2.0	1.6	1.9	2.0	1.1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	46.7	58.8	67.2	79.3	83.5	86.7	81.5	71.9	59.8	52.9	48.0	(19)	(20)
		MCWB	41.1	39.2	46.1	51.5	59.7	61.5	65.4	64.5	60.0	53.1	49.4	45.6	(20)	(21)
	2%	DB	42.4	42.8	52.5	62.5	73.3	78.0	81.8	77.9	68.2	56.7	50.2	45.5	(21)	(22)
		MCWB	39.4	38.2	42.7	48.8	56.5	59.7	64.3	63.8	58.0	52.0	47.6	43.3	(22)	(23)
	5%	DB	40.0	40.6	47.6	58.0	69.1	73.6	77.9	74.4	65.3	54.9	47.6	42.8	(23)	(24)
		MCWB	37.4	37.3	39.3	46.5	55.2	58.2	63.3	62.1	56.6	51.0	45.1	40.7	(24)	(25)
	10%	DB	37.9	38.6	44.0	54.2	65.0	69.8	74.4	71.0	62.7	52.9	45.4	40.3	(25)	(26)
		MCWB	35.7	35.9	37.4	44.4	53.3	56.2	61.4	60.3	55.5	49.6	43.2	38.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	41.8	47.1	53.5	62.6	64.5	68.2	67.3	62.8	55.7	50.9	46.4	(27)	(28)
		MCDB	43.6	44.5	57.6	64.0	74.5	76.2	79.7	75.4	67.7	57.5	52.0	47.4	(28)	(29)
	2%	WB	40.2	39.7	43.2	50.0	59.2	62.2	66.3	65.3	60.5	53.7	48.0	43.7	(29)	(30)
		MCDB	41.7	41.5	50.6	58.8	70.1	73.0	77.3	73.5	65.1	55.6	49.7	45.2	(30)	(31)
	5%	WB	38.1	38.1	40.8	47.7	56.6	60.2	64.6	63.6	58.6	52.0	45.6	41.0	(31)	(32)
		MCDB	39.6	40.0	45.4	54.6	65.7	70.1	74.4	71.6	63.0	54.2	47.2	42.4	(32)	(33)
	10%	WB	35.9	36.2	38.8	45.7	54.4	58.3	63.1	62.0	57.0	50.1	43.5	38.3	(33)	(34)
		MCDB	37.6	38.1	42.8	52.4	63.5	66.8	72.1	69.0	61.2	52.5	45.2	39.9	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.4	12.9	14.5	16.0	15.6	15.2	14.2	13.1	10.2	7.5	7.7	(35)	(36)
		MCDBR	9.1	11.3	18.9	21.6	22.4	22.3	21.3	18.6	16.9	12.4	10.7	9.8	(36)	(37)
	5% WB	MCWBR	7.9	8.0	11.1	11.5	10.9	9.8	8.9	8.0	9.3	8.6	9.5	8.9	(37)	(38)
		MCWBR	8.1	9.3	17.0	18.2	19.1	18.5	18.3	16.0	13.9	9.6	9.7	9.0	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub	taub	0.260	0.302	0.328	0.371	0.367	0.356	0.377	0.370	0.338	0.324	0.280	0.233	(39)	(40)
		taud	2.207	2.287	2.333	2.317	2.402	2.452	2.413	2.428	2.508	2.455	2.291	2.122	(40)	(41)
	Ebn at Noon	Ebn at Noon	167	223	255	262	273	278	268	261	252	215	159	128	(41)	(42)
		Edh at Noon	13	21	28	34	33	32	33	30	23	18	12	9	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg	RadAvg	91	284	754	1195	1553	1777	1588	1249	828	366	121	60	(43)	(44)
	RadStd	RadStd	17	52	81	162	159	132	205	99	95	71	33	6	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	+2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page